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**FIRST SEMESTER 2026-27**

**COURSE HANDOUT**

**Date: 25.07.2026**

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

**Course No : CHE G553**  
**Course Title : STATISTICAL THERMODYNAMICS**  
**Instructor-in-Charge : MOHIT GARG**

**1. Course Description:**

Review: Classical thermodynamic and elementary Statistical Mechanics, Macroscopic and microscopic descriptions of the state of a system, Equilibrium ensembles, the partition function and thermodynamic properties; System of independent particles; Fluctuations and the compressibility equation; Chemical equilibrium in ideal gas mixtures; Molecular based equations of state, SAFT, Lattice statistics; Real gases, Virial equation; The liquid state: lattice models, distribution functions theories, perturbation theories; Liquid mixtures: solution theories and local composition models, Statistical thermodynamics of electrolytes

**2. Scope and Objective of the Course:**

The objective of the course is to introduce the students with the microscopic description of the thermodynamics system and how macroscopic properties can be derived from microscopic description using statistical mechanics or statistical thermodynamics. In this course the statistical ensembles in classical mechanics, the concept of ensemble and phase space will help in understanding equilibrium statistical mechanics. Beyond this, this course will introduce the coding in either MATLAB or Python to do statistical analysis to find the thermodynamic or dynamic properties of the system. Several simulation techniques such as Monte Carlo Simulations and Molecular Dynamics using GROMACS and LAMMPS will be introduced to study applications of statistical thermodynamics to calculate various properties at nanoscale.

- Course learning outcomes:
  - a. Learning the fundamental concept of Statistical Thermodynamics
  - b. Learning the concept of molecular dynamics simulation and Monte Carlo Simulation
  - c. Learning the GROMACS and LAMMPS software to perform



### 3. Text Books:

TB 'Statistical Mechanics', Donald A McQuarrie, Viva Books Pvt. Ltd., 1st Indian Edn. (2003) (original: Harper & Row, New York, 1976).

### 4. Reference Books:

R1 'Introduction to modern statistical Mechanics', D. Chandler, Oxford University Press, NY (1987).

R2 'Introduction to Statistical Thermodynamics', Terrell L Hill, Addison Wesley (1960).

R3 Statistical Mechanics, Second Edition, R. K. Pathria, Butterworth-Heinemann (1996)

| S. No.                                                                                   | Module                                                | Lecture session                                                                                                                                                            | Reference              | Learning Outcome                                                                                                     |
|------------------------------------------------------------------------------------------|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------|
| R4: Will be shared, dynamically, in the classes. Students are advised to remain updated! |                                                       |                                                                                                                                                                            |                        |                                                                                                                      |
| 5. Course Plan                                                                           | 1. Introduction to Statistical Thermodynamics         | L.1.1. Concepts of chemical thermodynamics and need to study statistical thermodynamics                                                                                    | TB: Ch.-1<br>R1: Ch.-1 | <input type="checkbox"/> Understanding difference between Chemical Thermodynamics and Statistical Thermodynamics     |
|                                                                                          |                                                       | L.1.2 Introduction to the scope of statistical thermodynamics                                                                                                              | TB: Ch.1-2;            | <input type="checkbox"/> Scope and applications of Statistical Thermodynamics<br><input type="checkbox"/>            |
|                                                                                          |                                                       | L.1.3-L.1.4. Introduction to Probability, ensembles and most probable distribution                                                                                         | TB – Ch. 2             | <input type="checkbox"/> Understanding of Probability and its use in statistical mechanics                           |
|                                                                                          | 2. Ensemble Theory                                    | L.2.1 - L.2.3. Introducing the molecular partition function, discussion on various terms therein, interpretation of partition function and discussion on its applications. | TB: Ch. 2-3            | <input type="checkbox"/> Understanding microscopic description of a thermodynamic system<br><input type="checkbox"/> |
|                                                                                          |                                                       | L.2.4 – L.2.6. Canonical ensemble and derivation of partition function, Grand canonical and micro-canonical ensemble partition functions                                   | TB: Ch. 2-3, 7         |                                                                                                                      |
|                                                                                          | 3. Thermo-physical property calculations in ensembles | L.3.1 - L.3.2. Derivation of thermodynamic variables and thermodynamic properties from partition function                                                                  | TB: Ch.-3, 7           | <input type="checkbox"/> Understand how to derive thermodynamic properties from statistical mechanics                |
|                                                                                          |                                                       | L.3.3-L.3.4. Statistical explanation of second and third laws of thermodynamics                                                                                            | R4, R1                 | <input type="checkbox"/> Learn the statistical definition of entropy and its usefulness                              |
|                                                                                          | 4. Adsorption phenomenon                              | L.4.1 – L.4.3. Thermodynamic properties of perfect gas, Einstein and Debye theory of crystalline solids                                                                    | TB: Ch.-11;            | <input type="checkbox"/> Understand the microscopic view to adsorption phenomenon<br><input type="checkbox"/>        |
|                                                                                          |                                                       | L.4.4. Langmuir isotherms of adsorption of gas on lattice structure.                                                                                                       |                        |                                                                                                                      |



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|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|                                                                   |                                                                                                                                                                                            | BET isotherms of adsorption of gas on lattice structure. |                                                                                                                                                                  |  |
| 5. Phase Equilibrium                                              | L.5.1 – L.5.2. Ising model, Gibbs Ensemble (VLE), Thermodynamic Integration, Gibbs-Duhem Integration,                                                                                      | TB: Ch.-12, R3 Ch- 11                                    | <input type="checkbox"/> Understand the phase equilibrium from microscopic point of view<br><br><input type="checkbox"/>                                         |  |
|                                                                   | L.5.3-L.5.3. Free Energy calculation; Virial EOS: second virial coefficient                                                                                                                |                                                          |                                                                                                                                                                  |  |
| 6. Molecular Simulation in Grand Canonical Ensemble               | L.6.1 – L.6.2. Chemical equilibrium, Role of chemical potential and introduction to grand canonical Monte Carlo simulations.                                                               | R4                                                       | <input type="checkbox"/> Understand the role of chemical potential in adsorption phenomenon for various applications.<br><br><input type="checkbox"/>            |  |
|                                                                   | L.6.3-6.8. Running Monte Carlo simulation to study the role of chemical potential in sorption.                                                                                             | R4                                                       | <input type="checkbox"/> 1. Practice of software for Monte Carlo simulations.<br>2. Use of MATLAB or Python to compute statistical averages                      |  |
| 7. Molecular simulation in canonical and micro-canonical ensemble | L.7.1 – L.7.7. Introduction to polymer physics. Thermodynamics of mixing in polymer nanocomposites, Molecular simulation of polymers. Calculating ensemble average properties of polymers. | R4                                                       | <input type="checkbox"/> 1. Practice GROMACS software to run the Molecular simulations of polymers<br>2. Use of MATLAB or Python to compute statistical averages |  |
| 8. Calculating Statistical Averages                               | L.8.1 – L.8.3. Radial distribution function, Mean square displacement, Polymer, End to end distance, Radius of gyration, mean square fluctuations etc.                                     | R4                                                       | <input type="checkbox"/> Use of MATLAB or Python to compute statistical<br><br><input type="checkbox"/> averages                                                 |  |
|                                                                   | L.8.4. Review of recent problems                                                                                                                                                           | R4                                                       | <input type="checkbox"/> Keeping updated with the latest developments within this module                                                                         |  |

**6. Evaluation Scheme:**

| Component                | Duration (minutes) | Weightage (%) | Date & Time           | Nature of component (Close Book/ Open Book) |
|--------------------------|--------------------|---------------|-----------------------|---------------------------------------------|
| Mid-Semester Examination | 90                 | 25            | As announced by AUGSD | Closed and Open Book                        |



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|                                  |                                            |    |                                 |             |
|----------------------------------|--------------------------------------------|----|---------------------------------|-------------|
| Comprehensive Examination        | 180                                        | 35 | As announced by AUGSD           | Open Book   |
| Group Projects and Presentation  | In-Class & Take-home (10-15 min per group) | 20 | Date and time will be announced | Open Book   |
| Surprise Tests (best 2 out of 4) | 30 min each                                | 10 | -                               | Closed book |
| Assignment                       | Take home (1 week)                         | 10 | Announced                       | Open book   |

**7. Chamber Consultation Hour:** To be announced in the class.

**8. Notices:** All the notices would be put up on the Google Classroom.

**9. Make-up Policy:** Make-up request for the mid-semester examination is to be submitted 7 days in advance of the exam (or within 72 hrs. in case of medical emergency). For comprehensive examination the request is often made to AUGSD.

- It is mandatory to submit the necessary supporting documents (such as: listed below) along with make-up request:
  - o Letter of invitation/selection for a national/international event where the applicant is representing BITS Pilani.
  - o Proof of hospitalization, medical certificate, prescription, medicine purchase bill, etc.
- There is no makeup for the project presentations. However, students can mutually change the dates with permission from I/C.

**10. Note (if any):**

- Students have to adhere to the class timings and participate in class room discussion.
- There is no marks for class attendance. Attending classes would certainly make difference in learning & marks would reflect the same.

**Instructor-in-Charge**  
**Course No. CHE G553**



**General Handout (Part I):**

Introduction of Statistical Thermodynamics, Boltzmann Factor and Partition Functions, Concept of Ensembles, Canonical Partition Function and Ensemble, Partition functions and Ideal gases, Applications of Statistical Thermodynamics: i) Thermo-physical property calculations in ensembles, ii) Chemical Reactions in Ideal gases, Grand Canonical Partition Function and Ensemble, Micro Canonical Partition Function and Ensemble, Other ensembles, Ising model, Phase Equilibria, Gibbs Ensemble (VLE), Thermodynamic Integration, Gibbs-Duhem Integration, Free Energy calculation; Virial EOS: second virial coefficient; Intermolecular Potential Energy functions; Distribution functions: Radial Distribution Function, (RDF); Intermolecular Potential, Molecular Simulation Techniques: Molecular Dynamics (MD) Simulation, MD Simulation in various ensembles, Special Applications: From Surface Adsorption: Adsorption Isotherm, Diffusion of Porous Media.

**Prerequisite: NA**

**Corequisite: NA**

**SLO Mapping:**

| S. No. | Description                                                                                                     | SLO     |
|--------|-----------------------------------------------------------------------------------------------------------------|---------|
| CLO 1  | Learn how chemical engineering thermodynamics depends on the fundamental concepts of statistical thermodynamics | 1 - 21  |
| CLO 2  | Learn how statistical thermodynamics is applied in research and industry                                        | 22 - 29 |
| CLO 3  | Learn how various applications of statistical thermodynamics in research and industry are achieved              | 29 - 32 |